



भारत सरकार / Government of India
कृषि एवं किसान कल्याण मंत्रालय / Ministry of Agriculture & Farmers Welfare
कृषि एवं किसान कल्याण विभाग
Department of Agriculture & Farmers Welfare
वनस्पति संरक्षण, संगरोध और संग्रह निदेशालय
Directorate of Plant Protection, Quarantine & Storage
केंद्रीय कीटनाशी बोर्ड एवं पंजीकरण समिति
Central Insecticide Board & Registration Committee
एन. एच. - 4, फरीदाबाद 001 121 – (हरियाणा)
N.H.-IV, Faridabad-121 001 (Haryana)

MAJOR USES OF PESTICIDES

(Registered under the Insecticides Act, 1968)

(As on – 31/03/2026)

(Based on certificate issued)

**Disclaimer: 'The document has been compiled on the basis of available information for guidance and not for legal purposes'.*

BIO-INSECTICIDES

1. Major uses of Bio-insecticides - (Page No. 2 to 15).
2. Public health use - (Page No.15 to 18).

1. MAJOR USES OF BIO-INSECTICIDES					
Name of crop	Name of Insect	Dose/ha		Dilution in water (liter/ha)	Waiting period (Days)
		a.i. (g)	Formulation (g/ml)/%		
Azadirachtin 0.15% EC w/w Min. Neem Seed Kernel Based					
Cotton	White fly, Bollworms	-	2500-5000	500-1000	05
Rice (Paddy)	Thrips, Stem borer, Brown plant hopper, Leaf folder	-	1500-2500	500	05
Azadirachtin 00.30% EC (3000 PPM) Min. Neem Seed Kernel Based					
Cotton	American bollworm	-	4000	1000	05
Azadirachtin 01.00% (10000 ppm) EC Min. Neem Based					
Tea	Thrips	-	400-500	450	01
	Red spider mites	-	400-500	600	01
Azadirachtin 01.00% EC (10000 PPM) Min. Neem Based					
Tomato	Fruit borer (<i>Helicoverpa armigera</i>)	-	1000-1500	500	03
Brinjal	Shoot & fruit borer (<i>Leucinodes orbonalis</i>)	-	1000-1500	500	03
Azadirachtin 00.03% EC Min. Neem Oil Based					
Cotton	Bollworm (<i>Helicoverpa armigera</i>), Aphids	-	2500-5000	500	05
Rice (Paddy)	Leaf roller, Stem borer, Brown plant hopper	-	2000	1000	05
Azadirachtin 00.03% WSP (300 PPM) Neem Oil Based					
Bengal Gram (Gram or Chickpea)	Pod borer (<i>Helicoverpa armigera</i>)	-	-	-	07
Red Gram (Tur or Arhar)	Pod borer (<i>Melanagromyza</i> sp.)	-	2500-5000	500-1000	07

Cotton	Aphids, Jassids, Whitefly, Bollworms	-	2500-5000	500-1000	07
Okra (Bhindi)	Fruit borer, Whitefly, Leaf Hopper	-	2500-5000	500-1000	07
Brinjal	Shoot & Fruit borer, beetles	-	2500-5000	500-1000	07
Cabbage	Aphids, Diamond back moth, Cabbage worm, Cabbage looper	-	2500-5000	500-1000	07
Jute	Semi looper, Hairy caterpillar	-	2500-5000	500-1000	07
Azadirachtin 05.00% w/w Min. Neem Extract Concentrates					
Tea	Caterpillar, Pink mite, Red spider mites, Thrips	-	200.0	400	05
Tobacco	Tobacco caterpillar, Aphids	-	200.0	400	05
Rice (Paddy)	Brown plant hopper, Leaf folder, Stem borer	-	200.0	400	05
Cotton	Whitefly, Leaf hoppers, <i>Helicoverpa armigera</i> , Aphids	-	375.0	750	05
Cauliflower	<i>Spodoptera</i> , Diamond back moth, Aphids	-	200.0	400	05
Bhindi (Okra)	Leafhopper, whitefly, Aphid, Pod borer	-	200.0	400	05
Tomato	Aphids, Whitefly, Fruit borer	-	200.0	400	05
<i>Bacillus thuringiensis</i> var. <i>galleriae</i> 1593 M serotype H 59 5b, 1.3% flowable concentrate Potency 1500 IU/mg					
Cabbage & Cauliflower	Diamond back moth (<i>Plutella xylostella</i>)	-	06-1.0	500	-
Tomato	Fruit borer (<i>Helicoverpa armigera</i>)	-	1.0-1.5	500	-
Bhindi	Fruit borer (<i>Earias</i> spp.)	-	1.0-1.5	500	-

(Okra)					
Chilli	Fruit borer (<i>Spodoptera litura</i>)	-	1.5-2.0	1000	-
Cotton	Bollworm (<i>Helicoverpa armigera</i>)	-	2.0-2.5	1000	-
Rice (Paddy)	Leaf folder (<i>Cnaphalocrocis medinalis</i>)	-	1.0-3.0	1000	-
<i>Bacillus thuringiensis</i> var. <i>kurstaki</i>					
Cotton	Bollworm	-	750-1000	750-1000	-
<i>Bacillus thuringiensis</i> var. <i>kurstaki</i>, serotype H-39, 3B, Strain Z-52					
Cotton	Bollworms, <i>Spodoptera</i>	0.75-1.00	500-750	-	-
Rice (Paddy)	Stem borer & Leaf folder	1.50	500-750	-	-
Gram	<i>Heliothis</i> sp.	0.75	500-750	-	-
Pigeon Pea	<i>Heliothis</i> sp.	0.75	500-750	-	-
Soybean	<i>Spodoptera</i> , <i>Heliothis</i> , <i>Spilosoma</i> , Semilooper, Leaf miner	0.75	500-750	-	-
Tobacco	<i>Spodoptera</i> , <i>Heliothis</i>	1.50-2.00	500-750	-	-
Castor	Hairy caterpillar, <i>Achaea janata</i>	1.00	500-750	-	-
Teak	Defoliator (<i>Hyblaea pueria</i>), Skeletonizer (<i>Eutectona machaeralis</i>)	0.25-0.50	500-750	-	-
<i>Bacillus thuringiensis</i> serovar <i>kurstaki</i> (3a, 3b, 3c) 5.0% WP Potency 55000 SU (<i>Spodoptera</i> unit based) (5x10⁷ spore/mg)					
Cotton	American Bollworm	25.00-50.00	500-1000	500-1000	-
	Spotted Bollworm	37.50-50.00	750-1000	500-1000	-

Red gram	Pod Borer	50.00-62.50	1000-1250	500-1000	-
Cabbage	Diamond back moth	25.00-50.00	500-1000	500-1000	-
<i>Bacillus thuringiensis</i> var. <i>kurstaki</i> 0.5% WP serotype 3a, 3b, 3c, Strain DOR Bt-1, Potency 9000 IU/mg min. U/s 9(3b)					
Caster	Caster 5rustaki5r (<i>Achaea janata</i>)	-	0.25	250-300	-
<i>Bacillus thuringiensis</i> var. <i>kurstaki</i> 0.5% WP serotype 3a, 3b, 3c, Strain DOR Bt-1 NAIMCC-B-01118, Potency 13329 IU/mg min. U/s 9(3b)					
Pigeon pea	Bollworm (<i>Helicoverpa armigera</i>)	-	1-1.25	1000	-
<i>Bacillus thuringiensis</i> var. <i>kurstaki</i> 0.5% WP serotype 3a, 3b, 3c, Strain DOR Bt-1, Potency 9000 IU/mg min. U/s 9(3b)					
Caster	Caster 5rustaki5r (<i>Achaea janata</i>)	-	0.25- 0.375	250	-
<i>Bacillus thuringiensis</i> var. <i>kurstaki</i> 0.5% WP serotype 3a, 3b, 3c, Strain DOR Bt-1, Potency 16000IU/mg min.					
Chickpea	Chick pea pod borer (<i>Helicoverpa armigera</i>)	-	2.0	500	-
<i>Bacillus thuringiensis</i> var. <i>kurstaki</i> 2.5% AS (Spicbio-BTK AS)					
Gram	Gram pod borer (<i>Helicoverpa armigera</i>)	-	1.0-1.5	500	-
<i>Bacillus thuringiensis</i> var. <i>kurstaki</i> , Serotype H-3a, 3b, Strain Z-52 Potency:- 3000 IU/mg min – on Gypsy moth 32000 IU/mg min – <i>Trichoplusia vi</i> 50000 IU/mg min – <i>Helicoverpa armigera</i> 55000 IU/mg min – <i>Spodoptera exiqua</i>					
Cotton	Bollworms, Spodoptera	-	0.75-1.0 kg.	500-750	-
Rice	Stem borer & Leaf folder	-	1.50 kg.	500-750	-
Gram	<i>Helicoverpa armiger</i>	-	0.75 kg.	500-750	-
Pigeon Pea	<i>Helicoverpa armiger</i>	-	0.75 kg.	500-750	-

Soyabean	<i>Spodoptera litura</i> , <i>Helicoverpa armigera</i> , <i>Spilosoma brustak</i> , Semilooper, Leaf miner	-	0.75 kg.	500-750	-
Tobacco	<i>Spodoptera litura</i> , <i>Helicoverpa armigera</i>	-	1.50-2.00 kg.	500-750	-
Castor	Hairy caterpillar, Caster semilooper (<i>Achaea janata</i>)	-	1.00 kg.	500-750	-
Teak	Defoliator (<i>Hyblaea puera</i>), Skeletonizer (<i>Eutectona machaeralis</i>)	-	0.25-0.50% Sol.	As required.	-
<i>Bacillus thuringiensis</i> var. <i>kurstaki</i> Strain HD-1, serotype 3a, 3b, 3.5% ES for Import &repack. Potency 17600 IU/mg					
Cotton	Bollworms	-	750-1000	750-1000	-
Tea	Leaf roller (<i>Caloptilia theivora</i>)	-	1000	750-100	-
<i>Bacillus thuringiensis</i> var. <i>kurstaki</i> Serotype 3a, 3b, SA II WG Potency:- 53000 SU/mg, 32000 IU/mg					
Cabbage, Cauliflower	Diamond back moth	-	0.5 kg.	500-700	-
<i>Bacillus thuringiensis</i> var. <i>kurstaki</i> 10% WSL (CFU: 2x10⁹ /gm min.) Strain- NBAIR-BtG4, Accession no- JN120763, JN120765, Potency- 14245 IU/ml min.					
Pigeon Pea	Pod borer (<i>Helicoverpa armigera</i>)	-	10 liter	200	12 hrs
<i>Bacillus firmus</i> 1.5% AS (CFU count: 1 × 10 ⁸ /ml Min.) Strain name: AMMFA-BF1, NCMR Accession number MCC 0584)					
Tomato	Root knot nematode (<i>Meloidogyne incognita</i>)	--	5000	500	-
<i>Beauveria bassiana</i> 1.15% WP					
Cotton	Bollworms	-	400	750-1000	-
<i>Beauveria bassiana</i> 01.15% WP					
Cotton	Bollworm	-	2000	400	-
Rice (Paddy)	Leaf folder	-	2.50 kg/ha	750-850	-

<i>Beauveria bassiana</i> 1.15% WP. (1x10⁸ /gm min) Strain BB-ICAR-RJP, Accession No – MCC 1022					
Rice	Rice leaf folder (<i>Cnaphalocrocis medinalis</i>)	-	2.5 kg.	750-850	
<i>Beauveria bassiana</i> 1.15% WP (Strain : BB – 5372, own R & D Isolate)					
Rice	Rice leaf folder (<i>Cnaphalocrocis medinalis</i>)	-	2.5 kg.	600-750	-
<i>Beauveria bassiana</i> 1.15% WP (1x10⁸ /gm min) Strain ICAR, Research Complex, Umiam, Meghalaya, Accession No – NAIMCC-F-03045					
Rice	Rice leaf folder (<i>Cnaphalocrocis medinalis</i>)	-	2.5 kg.	750-850	-
<i>Beauveria bassiana</i> 1.15% WP (1x10⁸ /gm min) Accession No – NAIMCC-F-03045, Strain No. NBAIM, MAU.					
Rice	Rice leaf folder (<i>Cnaphalocrocis medinalis</i>)	-	2.5 kg/ha	750 liter/ha	-
<i>Beauveria bassiana</i> 1.15% WP (1x10⁸ /spores/ml) Strain BCRL, Accession No – BCRL Bbpx-6892					
Cabbage	Diamond back moth (<i>Plutella xylostella</i>)	1-1.5 litre	500-750	Apply using any type of sprayer (high, low or ultra low volume) which gives good coverage	NA
<i>Beauveria bassiana</i> 1.0% WP, Strain No: NBRI – 9947 (1x10⁸ CFU/gm Min.)					
Chick pea	Pod borer (<i>Helicoverpa armigera</i>)	-	3.0 kg.	500	-
<i>Beauveria bassiana</i> 1.0% WP (1x10⁹ CFU/gm min), Strain No. IPL/BB/MI/01					
Okra (Bhindi)	Fruit borer, Spotted bollworm	-	3.75-5.0 kg.	400-500	-
<i>Beauveria bassiana</i> 1.0% WP (1x10⁸ CFU/gm min), Strain No. SVBPU/CSP/Bb-10, Accession No. ITCC-7520					
Chick pea	Pod borer (<i>Helicoverpa armigera</i>)	-	3.0 kg.	500	-

<i>Beauveria bassiana</i> 5.0% WP, (1x10⁸ CFU/gm min) Strain IARI, Accession No. ITCC-7353					
Cabbage	Diamond back moth (<i>Plutella xylostella</i>)	-	2.0 kg.	500	-
<i>Beauveria bassiana</i> 5.0% SC, Strain: NBAII, Bangalore, Accession No. ITCC-7102, (Strain Isolated by Project Directorate of Bio-logical control, Bangalore)					
Tomato	Fruit borer (<i>Helicoverpa armigera</i>)	-	500	500	-
<i>Beauveria bassiana</i> 5.0% AS Strain: BB-AAU-RJP Accession No. MCC – 1024					
Tomato	Fruit borer (<i>Helicoverpa armigera</i>)	-	0.5	500	-
<i>Beauveria bassiana</i> 1.5% A.S. (CFU count: 1x10⁸ /ml Min.) Strain-AMMFA-BB1, NCMR Accession number MCC 0579					
Maize	Fall army worm (<i>Spodoptera frugiperda</i>)	-	3125	500	12 hrs
<i>Beauveria bassiana</i> 1.15% WP (1x10⁸ /gm min) Accession No – NAIMCC-F-03048					
Chick pea	Gram Pod Borer (<i>Helicoverpa armigera</i>)	-	2500	500	-
<i>Beauveria bassiana</i> 10.00% SC					
Cabbage	Diamond back moth	1-1.5	-	500-750	-
<i>Beauveria bassiana</i> 1.5% Liquid Formulation (CFU count 10X10⁸) Accession No.MTCC-5171					
Tomato	Fruit borer (<i>Helicoverpa armigera</i>)	2.0	Foliar spray	500	-
<i>Beauveria bassiana</i> 1.15% WP (1X10⁶ CFU /gm min)					
Cotton	Bollworm	20	-	400	-
<i>Lecanicillium (Verticillium) lecanii</i> 1.5% A.S. (CFU count: 1x10⁸ /ml Min.) Strain-AMMFA-V1, NCMR Accession number MCC 0577 (Biopesticide)					
Brinjal	Jassid, aphid, and white fly in brinjal.	-	125	500	-
<i>Metarhizium anisopliae</i> 1% WP (1x10⁸ CFU/gm min) Strain No. IPL/KC/44, Accession No. ITCC-6895.					
Brinjal	Shoot & Fruit borer (<i>Leucinodes orbonalis</i>)	-	2.5-5.0 Kg	500-750	-
<i>Metarhizium anisopliae</i> 1.15% WP (1x10⁸ CFU/gm min) Accession No. MTCC – 5173					

Rice	Brown plant hopper (<i>Nilapavata lugens</i>)	-	2.5 kg.	500	-
<i>Metarhizium anisopliae</i> 1.0% WP (1x10⁸ CFU/gm min) Strain No. IPL/KC/44 (Own R & D Isolate), Accession No. 6895.					
Brinjal	Shoot & Fruit borer (<i>Leucinodes orbonalis</i>)	-	2.5-5.0 kg	500-750	-
<i>Metarhizium anisopliae</i> 1.15% WP (1x10⁸ CFU/gm min) Strain No. AAI, Allahabad, Accession No. NAIMCC-F-03037.					
Chickpea	Heliothis armigera	-	2.5	500	--
<i>Metarhizium anisopliae</i> 1.5% Liquid Formulation (Accession No. MTCC-5173, T Stanes Ma-1 Strain)					
Paddy	Brown Plant Hopper (<i>Nilparvata lugens</i>)	-	2.0 lit	500	-
<i>Metarhizium anisopliae</i> 1.5% A.S. (CFU count: 1x10⁸ /ml Min.) Strain-AMMFAMA1, NCMR Accession number MCC 0578 (Bio-pesticide)					
Okra	Jassid, Aphid and White fly	--	3125	500	--
<i>Metarhizium anisopliae</i> 10% GR (CFU count 1 X 10⁸/gm. min.) Strain BCRL– Me, Accession No. ITCC 6911					
Crop	Common name of the target organism	Dosage /ha/application		Waiting period	
		Formulation	Method of application		
Potato	White grub	60kg.	Mix <i>Metarhizium anisopliae</i> (Grub –X 10% GR) with FYM @ 1 : 10 and apply at the root base by broadcasting method @ 6gm/m ² (=60 kg/ha) along the furrows at the time of sowing and next after one month at the time of earthing up.	NA	
<i>Pseudomonas fluorescens</i> 1.0% WP (Strain No. IIHR-PF-2, Accession No. ITCC- B0034)					
Tomato	Root-knot nematodes (<i>Meloidogyne</i> spp.)	Treat the seed with <i>Pseudomonas fluorescens</i> 1.0% WP @ 20 gm/kg of seeds & treat the nursery beds with the <i>Pseudomonas fluorescens</i> 1.0% WP @ 50 gm/sq.m and apply <i>Pseudomonas fluorescens</i> 1.0% WP @ 5 kg/ha enriched FYM* @ 5 tons/ha to the soil before transplanting.			

Brinjal	Root-knot nematodes (<i>Meloidogyne</i> spp.)	Treat the seed with <i>Pseudomonas fluorescens</i> 1.0% WP @ 20 gm/kg of seeds & treat the nursery beds with the <i>Pseudomonas fluorescens</i> 1.0% WP @ 50 gm/sq.m and apply <i>Pseudomonas fluorescens</i> 1.0% WP (@ 5 kg/ha enriched FYM* @ 5 tons/ha to the soil before transplanting.
Carrot	Root-knot nematodes (<i>Meloidogyne</i> spp.)	Treat the seed with <i>Pseudomonas fluorescens</i> 1.0% WP @ 20 gm/kg of seeds & treat the nursery beds with the <i>Pseudomonas fluorescens</i> 1.0% WP @ 50gm/sq.m and apply <i>Pseudomonas fluorescens</i> 1.0% WP (@ 5 kg/ha enriched FYM* @ 5 tons/ha to the soil before transplanting.
Okra	Root-knot nematodes (<i>Meloidogyne</i> spp.)	Treat the seed with <i>Pseudomonas fluorescens</i> 1.0% WP @ 20 gm/kg of seeds & treat the nursery beds with the <i>Pseudomonas fluorescens</i> 1.0% WP @ 50gm/sq.m and apply <i>Pseudomonas fluorescens</i> 1.0% WP @ 5 kg/ha enriched FYM* @ 5 tons/ha to the soil before transplanting.
<i>Trichoderma harzianum</i> 1.0% WP (Strain No. IHR-TH-2 Accessions No. ITCC 6888)		
Tomato	Root-knot nematodes (<i>Meloidogyne incognita</i>)	Treat the seeds with <i>Trichoderma harzianum</i> 1.0% WP @ 20 gm/kg of seeds & nursery beds with the <i>Trichoderma harzianum</i> 1.0% WP @ 50 gm/sq.m and also apply <i>Trichoderma harzianum</i> 1.0% WP (@ 5 kg/ha enriched FYM* @ 5 tons/ha to the soil before transplanting.
Brinjal	Root-knot nematodes (<i>Meloidogyne incognita</i>)	Treat the seeds with <i>Trichoderma harzianum</i> 1.0% WP @ 20 gm/kg of seeds & nursery beds with the <i>Trichoderma harzianum</i> 1.0% WP @ 50 gm/sq.m and also apply <i>Trichoderma harzianum</i> 1.0% WP (@ 5 kg/ha enriched FYM* @ 5 tons/ha to the soil before transplanting.
Carrot	Root-knot nematodes (<i>Meloidogyne incognita</i>)	Treat the seeds with <i>Trichoderma harzianum</i> 1.0% WP @ 20 gm/kg of seeds and apply <i>Trichoderma harzianum</i> 1.0% WP (@ 5 kg/ha enriched FYM* @ 5 tons/ha to the soil before sowing.
Okra	Root-knot nematodes (<i>Meloidogyne</i>	Treat the seeds with <i>Trichoderma harzianum</i> 1.0%
	<i>incognita</i>)	WP @ 20 gm/kg of seeds and apply <i>Trichoderma harzianum</i> 1.0% WP (@ 5 kg/ha enriched FYM* @ 5 tons/ha to the soil before sowing.
Gerbera	<i>Meloidogyne incognita</i>	Apply the Nemastin @ 50 gm/sq.m at the time of planting
Carnations	<i>Meloidogyne incognita</i>	Apply the Nemastin @ 50 gm/sq.m at the time of planting
Tuberose	<i>Meloidogyne incognita</i>	Apply 2 Kg Nemastin 1% Wp mixed in 2 tones of FYM per acre to the soil before planting
Banana	<i>Meloidogyne incognita</i>	Apply 2 Kg Nemastin enriched FYM @ 2 Kg/pant at the time of planting and at an interval of 3 months after planting for a period of one year
Acid lime	Citrus nematodes	Apply 2 Kg Nemastin enriched FYM @ 2 Kg/pant at the time of

	<i>(Tylenchulus semipenetrans)</i>	planting and at an interval of 3 months after planting for a period of one year			
Papaya	<i>Meloidogyne spp.</i> Reniform Nematodes <i>(Rotelenchulus reniformis)</i>	Apply 2 Kg Nemastin enriched FYM @ 2 Kg/pant at the time of planting and at an interval of 3 months after planting for a period of one year			
PB Rope L					
Crop	Common name of	Dosage			PHI (Days)
	pest	a.i.(mg/dispenser)	Pheromones dispensers	Recommended area of treatment in (Ha)	
Cotton	Pink bollworm <i>(Pectinophora gossypiella)</i>	>140	9875	25	NA
<i>Trichoderma harzianum</i> 1.5% WP (Strain No. IIHR-TV-5 Accessions No. ITCC 6889)					
Tomato	Root-knot nematodes (<i>Meloidogyne incognita</i>)	Treat the seed with <i>Trichoderma harzianum</i> 1.5% WP @ 20 gm/kg of seeds & treat the nursery beds with the <i>Trichoderma harzianum</i> 1.5% WP @ 50gm/sq.m and also apply <i>Trichoderma harzianum</i> 1.5% WP @ 5kg/ha enriched FYM* @ 5 tons/ha to the soil before transplanting.			
Brinjal	Root-knot nematodes (<i>Meloidogyne incognita</i>)	Treat the seed with <i>Trichoderma harzianum</i> 1.5% WP @ 20 gm/kg of seeds & treat the nursery beds with the <i>Trichoderma harzianum</i> 1.5% WP @ 50 gm/sq.m and also apply <i>Trichoderma harzianum</i> 1.5% WP @ 5 kg/ha enriched FYM* @ 5 tons/ha to the soil before transplanting.			
Carrot	Root-knot nematodes (<i>Meloidogyne incognita</i>)	Treat the seed with <i>Trichoderma harzianum</i> 1.5% WP @ 20 gm/kg of seeds & treat the nursery beds with the <i>Trichoderma harzianum</i> 1.5% WP @ 50 gm/sq.m and also apply <i>Trichoderma harzianum</i> 1.5% WP @ 5 kg/ha enriched FYM* @ 5 tons/ha to the soil before transplanting.			
Okra	Root-knot nematodes (<i>Meloidogyne incognita</i>)	Treat the seed with <i>Trichoderma harzianum</i> 1.5% WP @ 20 gm/kg of seeds & treat the nursery beds with the <i>Trichoderma harzianum</i> 1.5% WP @ 50 gm/sq.m and also apply <i>Trichoderma harzianum</i> 1.5% WP @ 5 kg/ha enriched FYM* @ 5 tons/ha to the soil before transplanting.			
<i>Trichoderma viride</i> 1.5% WP (Strain No. IIHR-TV-5 Accessions No. ITCC 6889)					
Tomato	Root-knot nematodes (<i>Meloidogyne incognita</i>)	Treat the seeds with <i>Trichoderma viride</i> 1.5% WP @ 20 gm/kg of seeds & nursery beds with the <i>Trichoderma viride</i> 1.5 % WP @ 50 gm/sq.m. and also apply <i>Trichoderma viride</i> 1.5% WP (@ 5 kg/ha enriched FYM*			

		@ 5 tons/ha to the soil before transplanting.			
Brinjal	Root-knot nematodes (<i>Meloidogyne incognita</i>)	Treat the seeds with <i>Trichoderma viride</i> 1.5% WP @ 20 gm/kg of seeds & nursery beds with the <i>Trichoderma viride</i> 1.5% WP @ 50 gm/sq.m. and also apply <i>Trichoderma viride</i> 1.5% WP (@ 5 kg/ha enriched FYM* @ 5 tons/ha to the soil before transplanting.			
Carrot	Root-knot nematodes (<i>Meloidogyne incognita</i>)	Treat the seeds with <i>Trichoderma viride</i> 1.5 % W P @ 20 gm/kg of seeds and apply <i>Trichoderma viride</i> 1.5% WP (@ 5 kg/ha enriched FYM* @ 5 tons/ha to the soil before Planting’.			
Okra	Root-knot nematodes (<i>Meloidogyne incognita</i>)	Treat the seeds with <i>Trichoderma viride</i> 1.5 % W P @ 20 gm/kg of seeds and apply <i>Trichoderma viride</i> 1.5% WP (@ 5 kg/ha enriched FYM* @ 5 tons/ha to the soil before Planting’.			
<i>Verticillium chlamydosporium</i> 1.0% WP, (2x10⁶ CFU/gm min) Strain – IIHR-VC-3 Accession No – ITCC-6898					
Tomato	Root-knot nematodes (<i>Meloidogyne incognita</i>)	Treat the seeds with <i>Verticillium chlamydosporium</i> 1.0% WP @ 20 gm/kg of seeds & nursery beds with the <i>Verticillium chlamydosporium</i> 1.0% WP @ 50 gm/sq.m and also apply <i>Verticillium chlamydosporium</i> 1.0% WP @ 5 kg/ha enriched FYM* @ 5 tons/ha to the soil before transplanting.			
Brinjal	Root-knot nematodes (<i>Meloidogyne incognita</i>)	Treat the seeds with <i>Verticillium chlamydosporium</i> 1.0% WP @ 20 gm/kg of seeds & nursery beds with the <i>Verticillium chlamydosporium</i> 1.0% WP @ 50 gm/sq.m and also apply <i>Verticillium chlamydosporium</i> 1.0% WP @ 5 kg/ha enriched FYM* @ 5 tons/ha to the soil before transplanting.			
Carrot	Root-knot nematodes (<i>Meloidogyne incognita</i>)	Treat the seeds with <i>Verticillium chlamydosporium</i> 1.0% WP @ 20 gm/kg of seeds and apply <i>Verticillium chlamydosporium</i> 1.0% WP @ 5 kg/ha enriched FYM* @ 5 tons/ha to the soil before transplanting.			
Okra	Root-knot nematodes (<i>Meloidogyne incognita</i>)	Treat the seeds with <i>Verticillium chlamydosporium</i> 1.0% WP @ 20 gm/kg of seeds and apply <i>Verticillium chlamydosporium</i> 1.0% WP @ 5 kg/ha enriched FYM * @ 5 tons/ha to the soil before transplanting.			
<i>Verticillium lecanii</i> 1.15%WP, (1x10⁸ CFU/gm min) Strain – AS MEGH-VL Accession No – MCC-1028					
Cotton	White flies	-	2500	500	-
Citrus	Mealybug (<i>Planococcus citri</i>)	-	2500	550	-

<i>Verticillium Lecanii</i> 1.50% Liquid Formulation, (1x10⁸ CFU/ml. min.) Strain – T Stanes VI-1, Accession No – MTCC-5172					
Tomato	White fly (<i>Bemisia tabaci</i>)	-	2000 (Foliar spray)	500	-
<i>Verticillium lecanii</i> 3.0% AS, (strain: Accession No. MCC-1127, Strain No. MPKV / Biocontrol/ RVN/ VL-01					
Onion	Thrips (<i>Thrips tabaci</i>)	-	2000-2500	500	-
<i>Verticillium lecanii</i> 5.0% SC, (Strain: Accession No. NFCCI - 2638					
Cabbage	Diamond Back Moth (<i>Plutella xylostella</i>)	-	500	500	-
<i>Verticillium lecanii</i> 5.0% SC, (1x10⁸ CFU/gm Min.) Strain – Own Red Isolate, Strain No. VI-17874, MTCC No.5716					
Rice	White backed plant hopper (<i>Sogatella furcifera</i>)	-	3.125 kg.	600	-
Nuclear Polyhedrosis Virus of <i>Helicoverpa armigera</i> 0.43% AS (1x10⁹ POB/ml)					
Cotton	<i>Helicoverpa armigera</i>	-	2700	400-600	-
Tomato	<i>Helicoverpa armigera</i>	-	1500	400-600	-
Nuclear Polyhedrosis Virus of <i>Helicoverpa armigera</i> 2.0% AS, Strain No. GBS/HNPV -01 (1x10⁹ POB/ml Min.)					
Pigeon pea	Pod borer (<i>Helicoverpa armigera</i>)	-	250-500	500-750	-
Gram	Pod borer (<i>Helicoverpa armigera</i>)	-	250-500	500-750	-
Nuclear Polyhedrosis Virus of <i>Helicoverpa armigera</i> 2.0% AS, Strain No. NBRI-8821 (1x10⁹ POB/ml Min.)					
Pigeon pea	Pod borer (<i>Helicoverpa armigera</i>)	-	500	500	-
Nuclear Polyhedrosis Virus of <i>Helicoverpa armigera</i> 2.0% AS, Strain No. IBH-17268 (1x10⁹ POB/ml Min.)					

Pigeon pea	Pod borer (<i>Helicoverpa armigera</i>)	-	250-500 ml	500-750	-
Gram	Pod borer (<i>Helicoverpa armigera</i>)	-	250-500 ml	500-750	-
Nuclear Polyhedrosis Virus of <i>Helicoverpa armigera</i> 2.0% AS, Strain No. BIL/HV-9 POB(1x109 POB/ml Min.)					
Pigeon pea	Pod borer (<i>Helicoverpa armigera</i>)	-	250-500	500-750	-
Chick pea	Pod borer (<i>Helicoverpa armigera</i>)	-	250-500	500-750	-
Tomato	Pod borer (<i>Helicoverpa armigera</i>)	-	250-500	500	-
Nuclear Polyhedrosis Virus of <i>Helicoverpa armigera</i> 2.0% AS, Strain No. IBL-17268					
Pigeon pea	Pod borer (<i>Helicoverpa armigera</i>)	-	250-500	500-750	-
Chick pea	Pod borer (<i>Helicoverpa armigera</i>)	-	500-1000	500-750	-
Nuclear Polyhedrosis Virus of <i>Helicoverpa armigera</i> 0.43% AS, Strain No. BIL/HV-9 (1x109 POB/ml Min.)					
Cotton	<i>Helicoverpa armigera</i>	-	2700	400-600	-
Tomato	<i>Helicoverpa armigera</i>	-	1500	400-600	-
Nuclear Polyhedrosis Virus of <i>Spodoptera litura</i> 0.5% AS, (1x109 POB/ml Min.)					
Tobacco	<i>Spodoptera litura</i>	-	1500	400-600	-
NPV of <i>Helicoverpa armigera</i> 0.5%AS, (1x109 POB/ml Min.)					
Chickpea	Pod borer (<i>Helicoverpa armigera</i>)	-	250	500	-
NPV of <i>Helicoverpa armigera</i> 2.0%AS, (1x109 POBs count / ml min) Biological Insecticide					
Chickpea	Pod borer	-	250	600	-

	(<i>Helicoverpa armigera</i>)				
<i>Paecilomyces lilacinus</i> 01.15% WP (Accession No. MTCC No. 5175, T-Stanes PI-1 Strain)					
Brinjal	Root Knot Nematode	03.0 kg	500 kg Organic manure/ Organic fertilizer	-	-
<i>Paecilomyces lilacinus</i> 1.15% WP (CFU 1¹⁰⁸ / gm. Min) (T Stanes –PI-1 Strain Accession No. MTCC 5175)					
Chilli	Root Knot Nematode	11.5 g	5.0 – 6.0	-	-
Cucumber	Root Knot Nematode	11.5 g	5.0 -6.0	-	-
<i>Paecilomyces lilacinus</i> 01.50% LF (CFU count 1x10⁸/ml min.)(Accession No. MTCC No. 5175, T-Stanes PI-1 Strain)					
Tomato	Root Knot Nematode (<i>Meloidogyne incognita</i>)	-	6000	500	-
2. PUBLIC HEALTH USE					
Name of Insect	Habitat	Dose		Surface	Waiting Period (days)
		a.i. (gm)	Formulation (gm)		
Azadirachtin 0.15% EC					
Mosquito larvae	Stagnant water, Drainage water, Puddle	1.0	1.0	10.7 m ²	-
	Iron containers, Machinery scraps, Iron box, Iron tanks	5.0	5.0	53.6 m ²	-
	Plastic scraps, Pit	933.3	933.3	01 ha	-
<i>Bacillus thuringiensis</i> var. israelensis WP					
Anopheles and Culex (larvae)	-	-	2-5 kg.	-	14-28
<i>Bacillus thuringiensis</i> var. israelensis , Serotype H-14 (VECTOBAC 12 AS) Potency 1200 ITU / MG (VCRC Serotype H-14 strain					
Culex	Drains, Cesspits Casuarina pits, Disused wells	-	5.0 litres	01 liter in 100 liter of water	-
Anopheles	Paddy fields, Ponds, Pools	-	10.0 litres	01 liter in 50 liter of water	-
Aedes	Tree holes, Disused tyres	-	10.0 litres	01 liter in 50 liter of water	-

Culex	Drains, Cesspits Casuarina pits, Disused wells	-	5.0 litres	01 liter in 100 liter of water	-
<i>Bacillus thuringiensis</i> var. israelensis, Serotyp H-14 (Vectobac 12 AS) potency 1200 ITU/mg					
Anopheles	Clean water, cement tanks	-	1-2 liters	-	-
Culex	Polluted water, Casspits, Cement tank, Stagnant and flowering drains	-	2-4 liters	Bacillus thuringiensis var. israelensis 5.0% A.S., [Strain VCRC B17], Accession No.: MTCC 5596-	-
<i>Bacillus thuringiensis</i> var. israelensis 5.0% AS (Visible spore count 8.6x10⁷ /ml min.) (Strain VCRC-B-17, Accession No.- MTCC 5596) potency 630 ITU/mg.min.					
Anopheles, Culex and Aedes	Highly polluted water	0.20 ml/m ²	1 liter	200liter	-
Anopheles, Culex and Aedes	Clean Water	0.10 ml/m ²	1 liter	200liter	-
<i>Bacillus thuringiensis</i> var. israelensis 5.0% AS (Strain VCRC-B-17, Serotype H-14, Accession No.- MTCC 5596) potency 630 ITU/mg.min.					
Culex	Polluted water (Drain, Cesspits, Casuarina, Pit, Disused well)	-	05-10 liters	01 liter in 50-100 liters of water	-
Anopheles	Clean water (Ponds, Pool, Paddy fields)	-	05 liters	01 liter in 100 liters of water	-
Aedes	Tree holes, disused tyres	-	10 liters	01 liter in 100 liters of water	-
Culex	Polluted water (Drain, Cesspits, Casuarina, Pit, Disused well)	-	10 lit (1 ml/m ²)	01 liter in 100 liters of water	-
Anopheles	Clean water (Ponds, Pool, Paddy fields)	-	5 liters(0.5 ml/m ²)	0.5 liter in 100 liters of water	-
Aedes	Tree holes, disused tyres	-	10 liters (1 ml/m ²)	01 liter in 100 liters of water	-
Culex	Drains, Cesspits, casuarinas pits, Disused Wells	-	5 lit/ha	01 liter in 100 liters of water	-
Anopheles	Paddy fields, ponds, pools	-	10 lit/ha	01 liter in 50 liters of water	-

Aedes	Tree holes, disused tyres	-	10 lit/ha	01 liter in 50 liters of water	-
Culex	Polluted water (Drains, Cesspits, Casuarina Pits, Disused wells)	10 lit (1 ml/m2		200 Liters of Water	-
Anopheles	Clean water (Paddy fields, Ponds, Pool)	5 (0.5 ml/m2)		200 Liters of Water	-
Bacillus thuringiensis var. israelensis (H-14) 5.0% AS					
Culex	Drains, Cesspits, casuarinas pits, Disused Wells	-	5 lit/ha	01 liter in 100 liters of water	-
Anoppheles	Paddy fields, ponds, pools	-	10 lit/ha	01 liter in 50 liters of water	-
Aedes	Tree holes, disused tyres	-	10 lit/ha	01 liter in 50 liters of water	-
Bacillus thuringiensis var. israelensis, Serotype H-14, 5% WP Potency 2000 ITU/mg					
Area and Breeding (Habitat)			Dose (g/m2)	Recommended application Frequency	
River bed pool			0.5	Weekly	
Cement tanks			0.5	Fortnightly	
Pokhars small kaccha or cement tanks with low walls			0.5	Weekly	
Pits and ditches			0.5	Weekly	
Paddy fields			0.5	Weekly	
Semi polluted pits			0.5	Weekly	
Ornamental fountains			0.5	Fortnightly	
Septic tanks			1.0	Weekly/Fortnightly	
Flood prone polluted cesspits and ditches			0.5	Weekly	
Drains with polluted stagnant or flowing very slowly			0.5	Weekly/Fortnightly	
Bacillus thuringiensis var. israelensis, Strain Designation- ABIL, Accession No. NAMICC-B01318 (CFU Count- 4.8 x 10⁸) Serotyp H-14, 5% WP Potency 7000 ITU/mg					
Name of Insect	Habitat	Formulation (lit/ha.)		Dilution in water (Liters)	Waiting period (Days)
		Gm/m2	Kg/ha		
Anopheles species, Culex species, Aedes	Clean water, (Cement tanks, coolers, Drains, Pools	0.75	7.50	200	-

	and Pits)				
species	Highly Polluted water- (Underground tanks, Container, Drums & Tyros)	1.00	10.00	200	-
<i>Bacillus thuringiensis</i> var. <i>sphaericus</i>1593 M serotype H 59 5b					
<i>Anopheles</i> species, <i>Culex</i> species	For Drains, Cesspits Cesspools, Paddy fields, ponds	-	112	1 liter in 10 liter of water	-
<i>Anopheles</i> species, <i>Culex</i> species	Casuarinas pits, unused wells, unused overhead tanks, Domestic wells (Not for drinking requirements)	-	112	1 liter in 10 liter of water	-
<i>Bacillus thuringiensis</i> var. <i>israelensis</i> 12% AS (Vectobac)					
<i>Anopheles</i> species	Clean water, cement tanks	-	1-2 liter	-	-
<i>Culex</i> species	Polluted water, cess pits, cement tanks, stagnant and flowing drains	-	2-4 liter	-	-
<i>Bacillus thuringiensis</i> var. <i>israelensis</i> 00.50% WP					
Mosquito spp.	Anopheles, <i>Culex</i> and <i>Aedes</i> (Habitat-Cement tank, Coolers, Drains, Pool pits, Highly polluted underground tanks, Container drums & Tyres.)	0.75 mg/m ²	-	200	-
<i>Bacillus thuringiensis</i> var. <i>israelensis</i> 05.00% WP					
Mosquito spp.	Anopheles, <i>Culex</i> and <i>Aedes</i> (Habitat-Cement Tank, Coolers, Drains, Pool pits	0.75 g/m ²	7.50 kg/ha.	200 L	-
	Highly polluted	1.00 g/m ²	10.00 kg/ha	200 L	-

	water (underground tanks, Container Drums and Tyres.)				
<i>Bacillus sphaericus</i> 1593 M serotype H 59 5b, 1.3% Flowable concentrate Potency 13000 IU/mg					
<i>Anopheles</i> species, <i>Culex</i> species	For Drains, Cesspits Cesspools, paddy fields, ponds	-	112 ml	1 liter/10 liter of water	-
<i>Anopheles</i> species, <i>Culex</i> species	Casuarinas pits, unused wells, unused overhead tanks, Domestic wells (Not for drinking requirements)	-	112 ml	1 liter/10 liter of water	-